



Optimal compatibility proportional screening of *Trichosanthis Pericarpium* - *Trichosanthis Radix* and its anti - Inflammatory components effect on experimental zebrafish and coughing mice

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ABSTRACT

Ethnopharmacological relevance: The herbal pair of *Trichosanthis Pericarpium* (TP) - *Trichosanthis Radix* (TR) can be seen in the famous formula “Beimu Gualou San”. It is a commonly selected combination of medicinal herbs for the treatment of cough with lung heat. Both drugs are derived from *Trichosanthes kirilowii* Maxim, a medicinal plant known for its ability to clear heat, resolve phlegm, produce saliva, and alleviate dryness. However, the optimal combination ratio and active ingredients of TP-TR have yet to be determined.

Aim of the study: This study aims to investigate the optimal combination ratio of TP-TR and its anti-inflammatory active ingredients in cough treatment.

Materials and methods: A zebrafish (*Danio rerio*) inflammatory injury model and response surface method were applied in the present study to determine the appropriate proportion of TP-TR. Chemical constituents in TP-TR were identified using HPLC-ELSD and UPLC-MS/MS methods. Subsequently, a cough mouse model was created using an ammonia solution to evaluate the effectiveness of the optimal TP-TR ratio. Network pharmacology and intestinal flora sequencing were used to validate the anti-inflammatory components of TP-TR.

Results: The herbal pair of TP - TR at the ratio of 1:2 showed an optimal anti-inflammatory effect, with a composite inflammatory factor score of 119.645 in the zebrafish experiment. TP-TR combination facilitated the dissolution of glutamine, inosine, cytosine, isoquercetin, and other substances. In the animal model, the TP-TR (1:2) treatment significantly reduced the frequency of coughs and prolonged cough latency compared to the model group. Results of the network pharmacology indicated that inflammatory-related factors such as TLR4, STAT3, EGFR, and AKT1 played crucial roles in cough treatment with TP-TR, consistent with the validation experiment. The 16s rDNA sequencing results revealed a significant increase in the abundance of *Clostridia_UCG-014*, *Lachnospiraceae*, *Christenella*, *Ruminococcus*, and other species in the intestinal tract of mice after modelling. TP-TR (1:2) reduced the abundance of pro-inflammatory flora such as *Clostridium_UCG-014* and *Lachnospira*, which were closely associated with L-lysine and trans-4-hydroxy-L-proline present in TP-TR according to correlation analysis.

Conclusion: TP-TR may promote the dissolution of glutamine, thymidine, inosine, cytosine, isoquercetin, and other components through their combination, thereby regulating the abundance of *Clostridium_UCG-014* and *Lachnospira* and exerting an antitussive effect. This study, for the first time, showed that TP-TR at a 1:2 ratio exhibits superior anti-inflammatory effects. In addition to inflammatory mediators like EGFR, TLR4, AKT1, and STAT3, gut microbes could also serve as potential regulatory targets of TP-TR in the treatment of cough. 2'-Deoxyguanosine monohydrate, L-lysine, L-leucine, γ -aminobutyric acid, L-valine, L-tryptophan, L-proline, trans-4-hydroxy-L-proline, L-methionine, uridine, 2'-deoxyinosine, guanosine, cucurbitacin B and cucurbitacin D were identified as its anti-inflammatory active ingredients.

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Abbreviations list*Abbreviations Full Name*

ACE	angiotensin converting enzyme	IL-8	interleukin-8
ADRB2	beta-2 adrenergic receptor	MET	mesenchymal - epithelial transition
AKT1	serine/threonine-protein kinase AKT	MMP1	matrix metalloproteinase
TLR4	Toll-like receptor 4	MPO	myelin myeloperoxidase
ALK	anaplastic lymphoma kinase	NOS1	nitric oxide synthase one
ALOX5	arachidonate 5-lipoxygenase gene	OPRD1	opioid receptor delta 1
CYP2D6	cytochrome P2D6	OPRM1	opioid receptor mu 1
EGFR	epidermal growth factor receptor	SLC6A2	solute carrier family 6 member 2
HPLC-ELSD	high performance liquid chromatography - evaporative light-scattering detector	STAT3	signal transducer and activator of transcription 3
IL-10	interleukin-10	TACR1	tachykinin Receptor 1
IL-13	interleukin-13	TERT	telomerase reverse transcriptase
IL-17	interleukin-17	TGF- β	transformation growth factor- β
IL-1 β	interleukin-1 β	TNF- α	tumor necrosis factor alpha
IL-4	interleukin-4	TRPA1	transient receptor potential ankyrin 1
IL-6	interleukin-6	TRPV1	transient receptor potential vanilloid 1
		UPLC-MS/MS	ultra performance liquid chromatography - tandem mass spectrometry
		VEGFA	vascular endothelial growth factor A

1. Introduction

Herbal pair is a special form of prescription compatibility, which is the smallest compatibility unit of a formula. The modern study of herbal pair is helpful to analyze the law of prescription composition, reveal the regularity of prescription formulation, and improve the effectiveness and safety of clinical drugs. The herbal pair of *Trichosanthis Pericarpium* - *Trichosanthis Radix* (TP-TR) can be seen in the famous formula “Beimu Gualou San” (*Yi-Xue-Xin-Wu*, Qing dynasty) for clearing heat, resolving phlegm, clearing lung, and calming asthma (Guo and Wang, 2023). In *Shi-Jin-Mo-Dui-Yao*, the herbal pair TP-TR is recorded as a commonly selected combination of medicinal herbs for the treatment of cough in modern Chinese medicine clinical practice. This simultaneous application of both drugs offers simplicity compared to traditional compound prescriptions, while still serving the benefits of compound compatibility. This approach not only improves effectiveness in the general practice but also reduces potential toxicities and adverse effects. TP is derived from the dried peel of *Trichosanthes kirilowii* Maxim, while TR is derived from its root. Both herbs are significant in their ability to clear heat from the body (Yu et al., 2018). In famous TCM doctor Shi Jinmo’s clinical prescriptions, combining TP and TR has proven to be highly effective in treating cough associated with lung heat, producing amplified therapeutic outcomes. Modern pharmacological studies have demonstrated that the n-butanol extract of TP can effectively inhibit the expression of inflammatory factors such as IL-6, IL-8, and TNF- α in BEAS-2B cells. TR, on the other hand, exhibits the ability to reduce NF- κ B activity and down-regulate pro-inflammatory factors (Wang et al., 2018). Nevertheless, further research is required to determine the optimal compatibility ratio of TP-TR and explore its active ingredients.

In Chinese medicine, cough is attributed to the invasion of the lungs by external pathogens or internal imbalances affecting the lung organ system. Coughs are categorized based on their underlying causes, such as wind-cold cough, wind-heat cough, phlegm-damp cough, and more. These classifications take into account factors like the nature of the pathogen and the presence of phlegm. On the other hand, modern medicine recognizes cough as a prominent symptom of respiratory diseases. It focuses on diagnosing and treating coughs by studying the physiological and pathological mechanisms of the respiratory system (Lucanska et al., 2020). Excessive coughing is frequently the result of intricate and diverse interactions involving nerve and immune cells within the lungs, vagal sensory ganglia, and the central nervous system. A shared characteristic among the different conditions associated with persistent coughing is the activation and sensitization of the coughing

center by inflammatory factors related to lung diseases (McGovern et al., 2018). Therefore, effectively alleviating cough requires a dual approach to reduce the release of the inflammatory mediators and suppress the cough center. Traditional Chinese medicine, such as *Fritillaria*, offers a relevant example. *Fritillaria* has the ability to down-regulate inflammatory mediators like IL-6, TNF- α , and others. It achieves the desired cough-relieving effect by inhibiting signaling pathways such as NF- κ B and mitogen-activated protein kinase (Zhang, H. et al., 2021). Drugs such as erdosteine are also used in treating cough based on their anti-inflammatory properties of reducing epithelial chemokines (Dal Negro, 2008).

In the present study here, we used a zebrafish (*Danio rerio*) inflammation model combined with the multi-indicator comprehensive index method to screen the appropriate compatibility proportion of TP - TR for the first time. In comparison to mammalian models, zebrafish embryos and juveniles serve as novel high-throughput drug screening models. Zebrafish possess a remarkably conserved immune system, enabling rapid immune responses to infections and tissue damage, making them an excellent model for screening anti-inflammatory drugs. Their ease of inducing inflammation further enhances their suitability for this purpose (LeBert and Huttenlocher, 2014).

To identify the anti-inflammatory active ingredients of TP-TR, a combination of network pharmacology and cough mouse models was employed. Network pharmacology enables the visualization of the “disease-drug-ingredient-target” network through the analysis of biological systems, facilitating the search for potential therapeutic compounds (Meng et al., 2020). In addition, lung disease is often associated with disturbances in the gut microbiota. Studies have found that dysbiosis, dominated by Firmicutes and Proteobacteria, is closely related to the expression of inflammatory factors in pulmonary leukocytes, and the up-regulation of the abundance of some Clostridium species can affect the production of anti-inflammatory regulatory T cells. The loss of T cells can promote the release of inflammatory mediators such as mast cells and basophils, thereby aggravating cough symptoms (Budden et al., 2017). It also has been reported that the abundance of *Coprobacillus*, *Clostridium ramosum* and *Clostridium hathewayi* in patients with lung disease is positively correlated with the severity of the respiratory symptoms (Zuo et al., 2020). Therefore, we conducted 16s rDNA sequencing analysis on intestinal microbes of cough mice after TP-TR intervention, and associated it with anti-inflammatory active ingredients. We can basically clarify the compatibility ratio and active ingredients of TP-TR, providing a reference for clinical medication.

2. Methods and materials

2.1. Medicinal herbs

Trichosanthis Pericarpium (lot No. 271220110) was purchased from Bozhou Zirui Pharmaceutical Co., LTD. Trichosanthis Radix (lot No. 210119) was purchased from Suzhou Tianling Chinese Medicine Decoction Pieces Co., Ltd. Professor Dong Ling, who is from the Institute of Horticulture, Anhui Academy of Agricultural Sciences, identified them as the dry mature peel and dry root of *Trichosanthes kirilowii* Maxim.

2.2. Animals

Zebrafish (*Danio rerio*) wild-type AB strain adult fish were purchased from Nanjing Yishu Lihua Biotechnology Co., Ltd. Adult zebrafish were reared in aquaculture equipment, culture water temperature: 28 °C, light/dark cycle: 14/10h, pH: 6.5–7.5, and brine shrimp were fed every day.

Specific Pathogen-Free (SPF) healthy ICR mice (Male, 4–6 weeks) weighing 18–22g were provided by Heathlake Laboratory Animal Co., Ltd., license number: SCKX (Shanghai) 2017 - 0005. Certificate number: 20170005061358. The animal experiments were approved by the Ethics Committee of Nanjing University of Chinese Medicine, ethics application number: 202111A026.

2.3. Chemical reagents

The reference compounds quercetin (lot No. HPS20151226), kaempferol (lot No. SNS20150828), rutin (lot No. RU20121202A), luteolin (lot No. MXCS20161208), isoquercetin (lot No. YHPS20150912) and apigenin (lot No. QCS20141104) were purchased from Nanjing Chunqiu Biological Co., Ltd. Luteolin (lot No. lw16012312), citrin (lot No. lw160111401), naringenin (lot No. lw200109100), cucurbitacin D (lot No. lw17090809), cucurbitacin B (lot No. lw16012602), cucurbitacin E (lot No. lw16101901), fructose (lot No. lw161129005), glucose (lot No. lw181128005), sucrose (lot No. lw180738024), raffinose (lot No. lw18012902) and stachyose (lot No. lw180309186) were purchased from Nanjing Liangwei Biotechnology Co., Ltd. Thymidine (lot No. 1000735425), hypoxanthine (lot No. A8626), uridine (lot No. U3750), adenosine (lot No. A9251), inosine (lot No. 086K1243), cytosine (lot No. SLBB2312V), cytidine (lot No. 0001446223), γ -aminobutyric acid (lot No. BCBD3661V), trans-4-hydroxy-L-proline (lot No. 1349920), L-glutamine (lot No. BCBC6452V) and L-citrulline (lot No. BCBC7694) were purchased from Sigma-Aldrich. Inc. 2'-Deoxyinosine (lot No. L1224040) was purchased from Aladdin Reagent Company. 2'-Deoxyguanosine (lot No. N07A7W12580) and guanine (lot No. Y09M8C30835) were purchased from Shanghai Yuanye Life Technology Co., Ltd. Phenylalanine (lot No. 20090316), L-valine (lot No. 20090318), L-proline (lot No. 20200522) and L-glutamine acid (lot No. 20090316) were purchased from Shanghai Huixing Biochemical Reagent Co., Ltd. L-serine (lot No. C11583738), L-threonine (lot No. C11584158), isoleucine (lot No. C11435950), L-bright amino acid (lot No. C11495324), L-tyrosine (lot No. C10874374), L-lysine (lot No. C10914803), L-methionine (lot No. C11385354) and L-tryptophan (lot No. C11385354) were purchased from Shanghai Macklin Biochemical Technology Co., Ltd. Pentoxifyverine (lot No. 19031012) was purchased from Sinopharm Rongsheng Pharmaceutical Co., Ltd. Ammonia (lot No. 14032630150) was purchased from Nanjing Chemical Reagent Co., Ltd. The purities of the reference compounds were all above 98%.

Trizol (lot No. MKCD9593) was purchased from Sigma-Aldrich. Inc. Isopropanol (lot No. 20200608) was purchased from Shanghai Alatin Biochemical Technology Co., Ltd. Chloroform (lot No. 20140925) was purchased from Shanghai Lingfeng Chemical Reagent Co., Ltd. Kit of EasyScript All in-one First-Strand cDNA Synthesis SuperMix, for qPCR

(One-Step gDNA Removal, lot No. P20329) and TransStart® Top Green qPCR SuperMix (lot No. O21220) was obtained from Beijing Transgen Biotechnology Co., Ltd.

2.4. Preparation of TP-TR herbal pair

Different proportions of TP - TR mixtures were prepared at ratios of 1:0, 0:1, 1:1, 1:2, 2:1, 1:3, 3:1, 2:3 and 3:2, each serving was 60 g. The mixed herbs were refluxed four times by adding water (1:8 w/v) for 1 h, water (1:6 w/v) for 1h, 80% ethnol (1:8 w/v) for 1 h and 80% ethnol (1:6 w/v) for 1 h, and then filtered through gauze. The four filtrates were combined and evaporated using rotary evaporation under vacuum at 60 °C to a certain volume to obtain extracts (1 g/mL) and subsequently used as a crude drug.

2.5. Optimal ratio screening by zebrafish inflammation model

The inflammation zebrafish model was established based on a previously described method with modifications (Lee et al., 2013). 3 dpf zebrafish were selected and transferred to 12-well plates, 5 strips/well. Then, 2 μ M hydrocortisone solution was added and soaked for 6 h. Acute injury was caused by severing the caudal fin with a sharp scalpel blade under a stereomicroscope. Six hours later, the tail cutting zebrafish were observed under a stereomicroscope. The successfully modeled zebrafish were selected and transferred to a new 12-well plate with 10 pieces per well. After aspirating and discarding the hydrocortisone solution, the nine different proportions of TP - TR extraction with different concentrations (25 μ g/mL, 50 μ g/mL, 100 μ g/mL, 200 μ g/mL, 400 μ g/mL, 800 μ g/mL) was placed respectively. RNA was extracted from zebrafish by Trizol and reversed transcription was performed. qPCR experiment was performed according to the TransGen qTOP Mix kit instructions. Quantitative fluorescence PCR (ABI-7500, American) was then used to determine the relative expression of IL-6, IL-8, IL-1 β , TNF- α , IL-17, IL-4, IL-10, IL-13, VEGFA, TGF- β in zebrafish. The primers are listed in supplementary materials Table S1.

Box-Behnken response surface method was employed to establish the quadratic multiple regression model fitting of TP-TR anti-inflammatory comprehensive score (Y) to the three factors A (TP), B (TR) and C (Drug concentration). The optimal ratio of TP-TR was predicted by response surface analysis.

2.6. Composition analysis of TP - TR

2.6.1. Determination of flavonoids and triterpenoids

Analyses were performed by using an ACQUITY™ UPLC System (Waters Corp., Milford, USA) coupled to a Xevo TQ-XS mass spectrometry (Waters Corp., Milford, USA). The separation was performed on an Acquity BEH C18 Column (100 mm $\times$ 2.1 mm, 1.7 μ m) held at 35 °C. The mobile phases, delivered at a flow rate of 0.4 mL/min by using a linear gradient program, were consisted of two solvent systems, including eluent A (water) and eluent B (methanol). The linear elution gradient program was as follows: 0–6.0 min, 10–40% B; 6–9 min, 40–95% B; 9–10 min, 95–97% B; 10–13 min, 97% B; 13–14 min, 97%–10%B. The injection volume was 2 μ L. The detection was operated under multiple reaction monitoring (MRM) mode. The MS parameters were set as follows: source temperature 150 °C, desolvation temperature 500 °C, capillary voltage 3 kV, desolvation gas 1000 L/h and cone gas 50 L/h. Parameters of flavonoids and triterpenes by mass spectrometry see Supplementary materials Table S2.

2.6.2. Determination of nucleoside and amino acid components

Analyses were performed by using an ACQUITY™ UPLC System (Waters Corp., Milford, USA) coupled to a Xevo TQ-XS mass spectrometry (Waters Corp., Milford, USA). The separation was performed on an Acquity UPLC BEH Amide Column (100 mm $\times$ 2.1 mm, 1.7 μ m) held at 35 °C. The mobile phases, delivered at a flow rate of 0.4 mL/min by

using a linear gradient program, were consisted of two solvent systems, including eluent A (5 mmol/L ammonium formate, 5 mmol/L ammonium acetate and 0.2% formic acid in water, v/v) and eluent B (1 mmol/L ammonium formate, 1 mmol/L ammonium acetate and 0.2% formic acid in acetonitrile, v/v). The linear elution gradient program was as follows: 0–9.0 min, 10–18% A; 9–15 min, 18–20% A; 15–18 min, 20–46% A; 18–19 min, 46% A; 19–20 min, 46%–10% A; The injection volume was 2 μ L. The MS parameters were set as follows: source temperature 150 °C, desolvation temperature 500 °C, capillary voltage 3 kV, desolvation gas 1000 L/h and cone gas 50 L/h. Parameters of nucleoside and amino acid components by mass spectrometry see Supplementary materials Table S3.

2.6.3. Determination of monosaccharides

Analyses were performed by using a 2695 HPLC System coupled to a 2424 Evaporative light scattering detector. The separation was performed on a Prevail Carbohydrate ES Column (250 mm $\times$ 4.6 mm, 5 μ m). The analysis was achieved with gradient elution using acetonitrile (A) and water (B) as the mobile phase at a flow rate of 1 mL/min with a temperature 30 °C. The gradient elution program: 0–7 min, 75% A; 7–17 min, 75%–55% A; 17–19 min, 55%–50% A; 19–21 min, 50%–75% A; 21–25 min, 75% A. The injection volume was 10 μ L.

2.7. Network pharmacological analysis

Reference the “Network Pharmacology Evaluation Method Guidance - Draft” (Li, 2021) and literatures method (Lin et al., 2021; Zhang, Z.-Q. et al., 2021). See Supplementary materials Text S1 for details.

2.8. Evaluation of antitussive effect of TP-TR

2.8.1. Cough mice model induced by ammonia water

Cough mice model induced by the concentrated ammonia water was established (Kuang et al., 2018). ICR mice were randomly divided into a normal control group (C), a model control group (M), a positive drug control group (Y), a TP low dose group (TPL), a TP high dose group (TPH), a TR low dose group (TRL), a TR high dose group (TRH), a TP - TR low dose group (TPTRL) and a TP - TR high dose group (TPTRH) with 12 mice in each group. The positive control group was given Pentoxifyrine (formed as 0.1% suspension) 50 mg/kg by gavage. The low-dose groups were given 3 g/kg of drug extract (clinical equivalent dose), and the high-dose groups were given 9 g/kg of drug extract. The model control group and normal control group were given equal volume of normal saline once a day for 7 days. One hour after the last administration, the mice were put into a 5 L glass cover, a cotton ball with 1 mL of 28% aqueous ammonia was then placed, and the lid was immediately closed. The cough latency of the mice (the time from putting the cotton ball in to the first cough of the mice) and the cough frequency within 3 min were observed and recorded. The cough suppression rate (%) was calculated.

After observation, all mice were anesthetized with isoflurane, and lung tissue and colon contents of mice were collected and sacrificed by cervical dislocation.

2.8.2. Detection of inflammatory cytokines expression by RT-qPCR

Trizol reagent was used to extract total RNA from sample in accordance with the manufacturer's protocol. Then, 2 μ g of total RNA was used as a template for reverse transcription to synthesize cDNA. Quantitative fluorescence PCR (ABI-7500, Thermo Fisher Scientific, USA) was then used to determine the relative expression of IL-6, IL-8, IL-1 β , TNF- α , IL-17, IL-4, IL-10, IL-13, VEGFA, TGF- β in zebrafish and EGFR, TLR4, STAT3, AKT1 in cough mice lung tissue. The primers are listed in supplementary materials Table S1.

2.9. Gut microbiota analysis

On the 6th day of administration, the feces of mice in each group were collected for intestinal flora analysis. The collected fecal samples were thawed on ice, fully mixed and centrifuged to extract genomic DNA, and the extracted genomic DNA was detected by 1% agarose gel electrophoresis. The primers were designed according to the conserved regions, and the ends of the primers were modified with sequencing linkers to amplify the bacterial v3-v4 region. The PCR products were recovered using a 2% agarose gel, and the PCR products were detected and quantified with the QuantiFluor™-ST blue fluorescence quantitative system (Promega Company, USA). The Miseq PE250 platform was used for sequencing (Shanghai Meiji Biomedical Technology Co., Ltd., China).

2.10. Processing of experimental results

The experimental data were processed by SPSS 12.0 statistical software, and the data were expressed as mean $\pm$ standard deviation (SD). One-way analysis of variance was used to compare the differences between groups. $P < 0.05$ indicated a significant difference and was considered statistically significant.

3. Results

3.1. The optimal proportion of TP-TR against inflammation

Using the relative expression levels of IL-6, IL-8, IL-1 β , TNF- α , IL-17, IL-4, IL-10, IL-13, VEGFA, and TGF- β in zebrafish (supplementary materials Table S4), we employed a multi-index comprehensive index method in conjunction with response surface analysis to assess the anti-inflammatory effect of the TP-TR herbal pair. All RQ (Relative Quantification) values were first normalized. Thereafter, weight coefficients for the standardized indicators were calculated using the expert scoring method. This involved independent evaluations by knowledgeable experts who assigned weight coefficients to each indicator based on its relative importance in the overall evaluation of the total effect. The weight coefficients were determined by considering the detection frequency statistics of the indicators. Additionally, we referred to relevant articles on zebrafish inflammation models from the past decade, which yielded a total of 1139 reports. Among these, IL-6 was detected in 178 studies, IL-8 in 76 studies, IL-1 β in 82 studies, TNF- α in 99 studies, and IL-17 in 94 studies. IL-10, IL-4 in 34, TGF- β in 38, IL-17 in 16, IL-13 in 20, and VEGFA in 8 papers. Combining the expert scoring method with the frequency statistics of each index detected in the literatures, the weight coefficients of IL-6, IL-8, IL-1 β , TNF- α , IL-10, IL-4, TGF- β , IL-17, IL-13, and VEGFA was found to be 20, 10, 10, 10, 10, 4, 4, 2, 2, and 1, respectively. According to the anti-inflammatory score of TP - TR (supplementary materials Table S5), quadratic regression fitting was performed using Design-Expert 12.0 software, and the fitting equation was set at $Y = 92.39 - 35.21A + 21.02B - 30.53C - 25.64AB + 0.4235AC - 6.33BC - 84.05B^2$, $P = 0.0037$ (supplementary materials Table S6). When the prediction ratio of TP - TR was 0.775 : 1.561 (nearly to 1 : 2), there was the best the anti-inflammatory effect (Fig. 1, supplementary materials Table S7).

3.2. Chemical interaction analysis of TP-TR

We have established the UPLC-MS/MS method for simultaneous determination of flavonoids and triterpenoids, nucleoside and amino acid components, and the HPLC-ELSD method for simultaneous determination of monosaccharides in *T. kirilowii* (Zhang, H.Q. et al., 2019a; Zhang, H.Q. et al., 2019b). These compounds contents in TP - TR (1 : 2) were analyzed here. The findings revealed that the herbal pair contained significant amounts of monosaccharides, particularly fructose (35.34 mg/g), glucose (27.32 mg/g), and stachyose (30.07 mg/g). The total

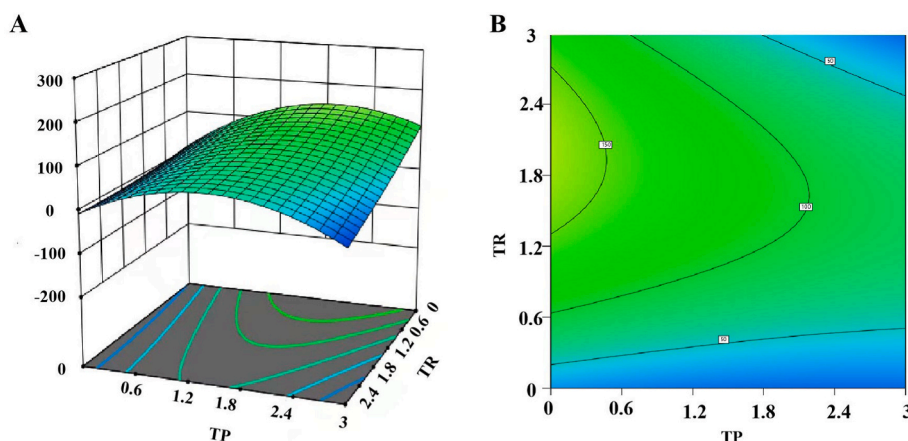


Fig. 1. Response surface plot of TP-TR herbal pair. (A) response surface diagram, (B) contour map.

amino acid content was determined to be 8854.34 $\mu\text{g/g}$, with citrulline being the most abundant amino acid (7901.47 $\mu\text{g/g}$). Moreover, the contents of rutin and cucurbitacin B were found relatively higher compared to other flavonoids and triterpenoids, measuring at 26.26 $\mu\text{g/g}$ and 450.97 $\mu\text{g/g}$, respectively (supplementary materials Table S8).

3.3. Network pharmacological results

3.3.1. Drug-to-active ingredient and candidate target intersection screening

A total of 132 constituents of TP and 76 of TR were collected from the TCMSF platform, literatures and our component analysis results. We gathered a combined number of 1346 potential targets by examining the 132 chemical constituents of TP and the 76 chemical constituents of TR. Subsequently, we used a Venn diagram to compare these constituent targets with 207 targets associated with cough-related conditions within 45 target genes were determined. These 45 targets correspond to 128 chemical constituents in TP-TR.

3.3.2. Active ingredient-potential target network construction

According to the screened active ingredients and related potential targets of TP and TR, Cytoscape software was used to establish a network map of active ingredients-potential targets (supplementary materials Fig. S1). In the figure, two red nodes represent the medicinal materials of TP-TR. Additionally, there were 128 purple nodes representing the 128 chemical components of the herbal pair, and 45 blue nodes representing potential targets associated with their active components. We calculated the average degree value and selected targets and compounds larger than the average degree value as the core targets and compounds (Li et al., 2023). Twenty targets with the highest degree values are found to be EGFR, NOS1, ALOX5, SLC6A2, TRPV1, MPO, AKT1, TERT, ACE, STAT3, MMP1, OPRM1, TRPA1, ALK, ADR β 2, TNF- α , MET, OPRD1, CYP2D6, TACR1. Moreover, the top ten compounds were luteolin, apigenin, cucurbitacin D, acenaphthylene, tangeretin, schottenol glucoside, gentic acid, diosmetin, cucurbitacin E, arvenin III. A higher degree value indicates a stronger correlation between the compound and cough, implying that the potential targets and compounds play a significant role in treating cough related to lung heat (supplementary materials Table S9).

3.3.3. PPI network construction

The obtained 45 potential targets of drugs on active ingredients were imported into the STRING database, and the direct and indirect interactions of the target proteins were obtained (Fig. 2). The findings revealed that the key targets for TP-TR in cough treatment primarily included IL-6, TNF- α , ALB, IL-1 β , TP53, EGFR, AKT1, CXCL8, TLR4, STAT3, among others. Inflammatory factors like IL-6 and TNF- α , when released, can lead to acute lung injury. In previous studies involving

transgenic animal models, as well as cellular and molecular biology investigations, have also demonstrated the significant involvement of EGFR in lung development, homeostasis, repair, and disease progression (Vallath et al., 2014).

3.3.4. Key target gene function analysis and metabolic pathway analysis

The Metascape database was utilized to perform gene and functional analysis on the 45 key targets. The results revealed that these targets were associated with various gene functions, such as membrane rafts, dendritic membranes, transferase complexes, nuclear membranes, neuronal projection membranes, and cytoplasmic perinuclear regions, which are closely linked to airway-related processes like inflammation, airway epithelial cell damage, airway hyperresponsiveness, and airway remodeling. The molecular functions identified a positive regulation of protein phosphorylation, positive regulation of peptidyl-serine phosphorylation, positive regulation of kinase activity, and positive regulation of transferase activity. Additionally, the biological processes involved in the protein kinase activity, protein phosphatase binding, and cytokine activity. The representative entries were selected based on their respective P-values (supplementary materials Fig. S2, Tables S10 and S11). Among these, cancer pathways involve AGTR1, AKT1, BRAF, CASP8, EGFR, ERBB2, IL-6, CXCL8, MET, MMP1, PIK3CA, MAP2K1, RET, STAT3, TGFBR1, TP53 and cytokines, while non-small cell lung cancer involves AKT1, ALK, BRAF, EGFR, ERBB2, PIK3CA, MAP2K1, STAT3, TP53 and other factors, HIF-1 signaling pathway AKT1, EGFR, ERBB2, HMOX1, IFNG, IL-6, PIK3CA, MAP2K1, STAT3, TLR4 and other factors. EGFR tyrosine kinase inhibitor resistance pathways involve AKT1, BRAF, EGFR, ERBB2, IL-6, MET, PIK3CA, MAP2K1, STAT3 and other factors.

Quercetin was reported to have immunomodulatory and bronchodilatory properties (Oliveira et al., 2015), while Luteolin was found to have antiallergic effects in mouse models of allergic asthma and rhinitis (Jang et al., 2017). β -Sitosterol has also been proved to have anti-asthmatic effects by inhibiting cellular responses and the release and synthesis of T2 cytokines (Mahajan and Mehta, 2011). By examining via the PPI network, gene function, and metabolic pathways, we hypothesized that the TP-TR herbal pair has the potential to modulate non-small cell lung cancer signaling through the targeting of IL-6, EGFR, TNF, IL-1 β , ALB, NOS1, AKT1. Moreover, pathways such as the HIF-1 signaling pathway, Toll-like receptor signaling pathway, IL-17 signaling pathway, and others are expected to contribute to the treatment of cough associated with lung heat.

3.3.5. Validation of anti-inflammatory targets by molecular docking

Based on the results of network pharmacology, molecular docking was performed to validate the binding modes of the top two active components (luteolin and cucurbitacin D) and the top four targets

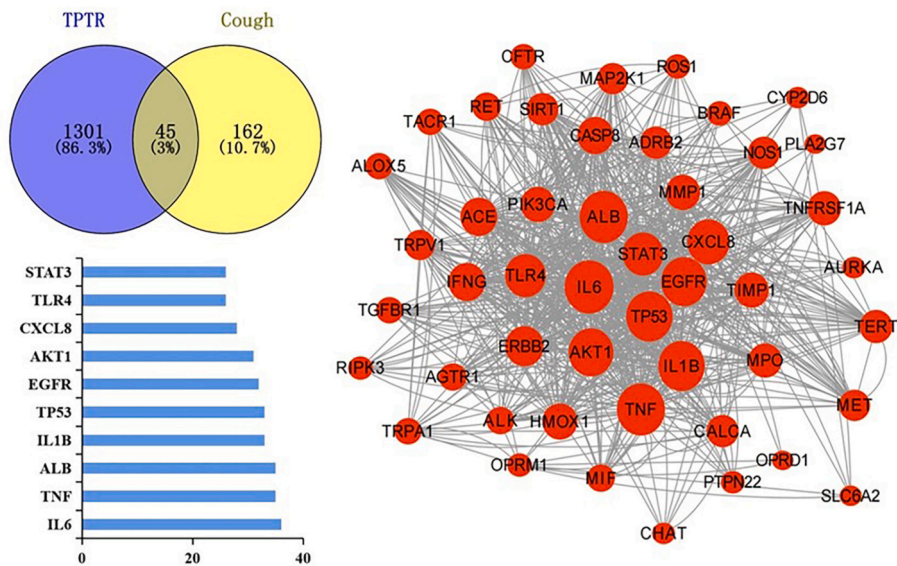


Fig. 2. PPI network diagram of TP-TR herbal pair for treating cough.

(AKT1, EGFR, STAT3 and TLR4). The crystal structure of the potential targets (receptors), including AKT1 (PDB ID: 4GV1, 1.49 Å), EGFR (PDB ID: 5UG9, 1.33 Å), STAT3 (PDB ID: 1BJ1, 2.40 Å), TLR4 (PDB ID: 3FXI, 3.10 Å) was downloaded through the PDB database. The 3D structures of two active ingredients (ligands) were obtained from TCMSP. Docking results showed that there were a strong binding affinity between the potential targets and active ingredients (binding energy ≤ -4.5 kcal/mol) (supplementary materials Fig. S3, Table S12), which preliminarily verified the core potential targets of TP-TR for cough treatment.

3.4. Anti - inflammatory effect of TP - TR in coughing mice

3.4.1. Antitussive effect of TP - TR

From the network pharmacology, we concluded that the potential targets of TP - TR in the treatment of cough were mainly IL-6, TNF, ALB, IL-1 β , TP53, EGFR, AKT1, CXCL8, TLR4, STAT3, while the signal pathway of non-small cell lung cancer, HIF-1, Toll-like receptor and so on involved in it. The cough mice induced by ammonia water were treated with the optimized TP-TR herbal pair (1:2) to evaluate its antitussive effect and observe changes in serum levels of IL-6, IL-1 β , and TNF- α . In comparison to the model group, the high dose of TP-TR group exhibited a significant reduction in cough frequency within 3 min ($P < 0.001$). The cough suppression rate of the high dose TP-TR group outperformed both the TP group and the TR group, achieving a rate of 65.71%. (Table 1).

The expression of AKT1, EGFR, STAT3 and TLR4 in mice lung tissue were examined by RT-qPCR. Compared with the control group, the mRNA expression of the target genes were found significantly increased ($P < 0.05$) in the model group. TLR4 exhibited the most significant

increase in expression, reaching a level 4.46 times higher than that of the control mice. When compared to the model group, the administration of TP-TR resulted in varying adjustment in the mRNA expression levels of AKT1, EGFR, STAT3, and TLR4. In the TP-TR low dose group, the expression levels of AKT1 and STAT3 decreased by 62.66% and 50.83% respectively. The TP-TR high dose group demonstrated the most effective correction of EGFR, with its expression decreasing by 61.61% after intervention. Among the administration groups, the TR low dose group showed the most significant reduction in TLR4 expression, with a decrease of 70.95% after administration. (Fig. 3).

3.4.2. Gut microbiota analysis

The species composition of the intestinal microbes of mice in each group was analyzed. According to the tendency of individual rarefaction curves (supplementary materials Fig. S4), it was revealed that sequencing data were reasonable and could reflect microbial information in all samples. The species diversity of the samples was evaluated at both OTU level and phylum level (supplementary materials Figs. S5–A). There were 835 OTUs in all experimental groups. Compared with the control mice, the OTUs species of model mice was significantly up-regulated, and 109 and 237 OTUs were found independent to the control group and the model group, respectively. Following the intervention, the composition of the intestinal microbial population in each group of mice underwent changes. Specifically, 133 OTUs were unique to the TP group, 72 OTUs were unique to the TR group, and 86 OTUs were unique to the TP-TR group. Among all the experimental groups, a total of 128 genera were identified, including 2 genera in the control group, 4 genera in the model group, 5 genera in the TP group, and 1 genus in the TR group. Analysis of the OTU classification and species

Table 1

The effect of TP-TR herbal pair on the cough mice induced by ammonia water. (Compared with the normal control group, $^{###}P < 0.001$. Compared with the model group, $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$.

Group	Dose (g/kg)	Incubation period (s)	cough frequency within 3 min (times)	Inhibition rate
Normal control	–	–	–	–
Model control	–	83.83 $\pm$ 24.31 $^{###}$	17.50 $\pm$ 3.45 $^{###}$	–
Positive drug control	0.05	118.00 $\pm$ 42.68 **	2.50 $\pm$ 1.05 ***	85.71%
TPL	3	109.67 $\pm$ 20.62	9.33 $\pm$ 3.33	46.67%
TPH	9	109.33 $\pm$ 26.87	8.83 $\pm$ 2.32 ***	49.52%
TRL	3	66.17 $\pm$ 33.52	15.17 $\pm$ 6.97	13.33%
TRH	9	86.17 $\pm$ 22.89	12.33 $\pm$ 3.14 *	29.52%
TPTRL	3	105.83 $\pm$ 39.00	7.00 $\pm$ 2.28	60.00%
TPTRH	9	100.00 $\pm$ 28.11	6.00 $\pm$ 3.46 ***	65.71%

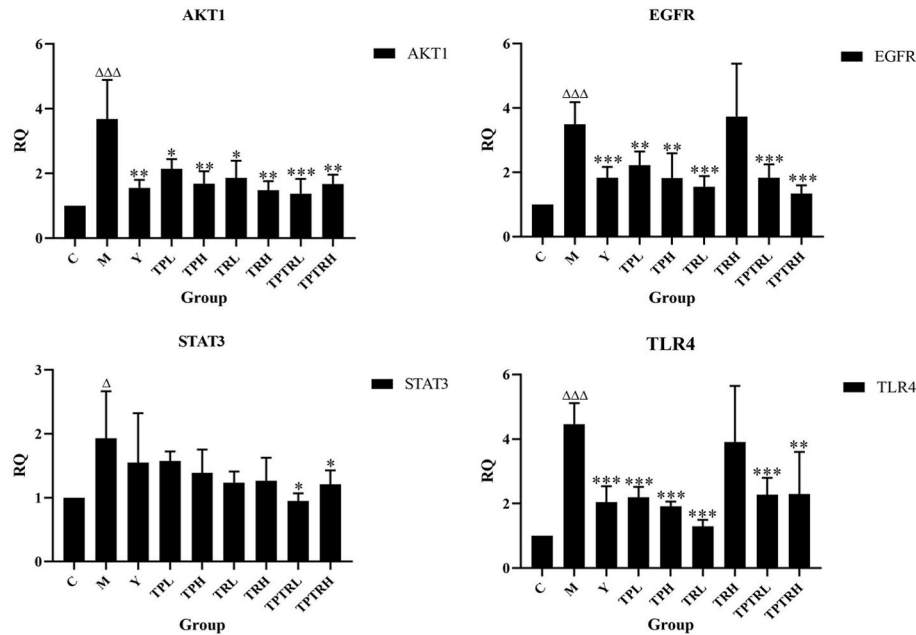


Fig. 3. Effects of TP-TR (1:2) on the expression of cytokines in lung tissue of cough mice (n = 6). C: the normal control group. M: the model control group. Y: the positive drug group. TPL: TP low dose group. TPH: TP high dose group. TRL: TR low dose group. TRH: TR high dose group. TPTRL:TP-TR low dose group. TPTRH: TP-TR high dose group. $\Delta P < 0.05$, $\Delta\Delta\Delta P < 0.001$, vs. the normal control group. $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, vs. the model control group.

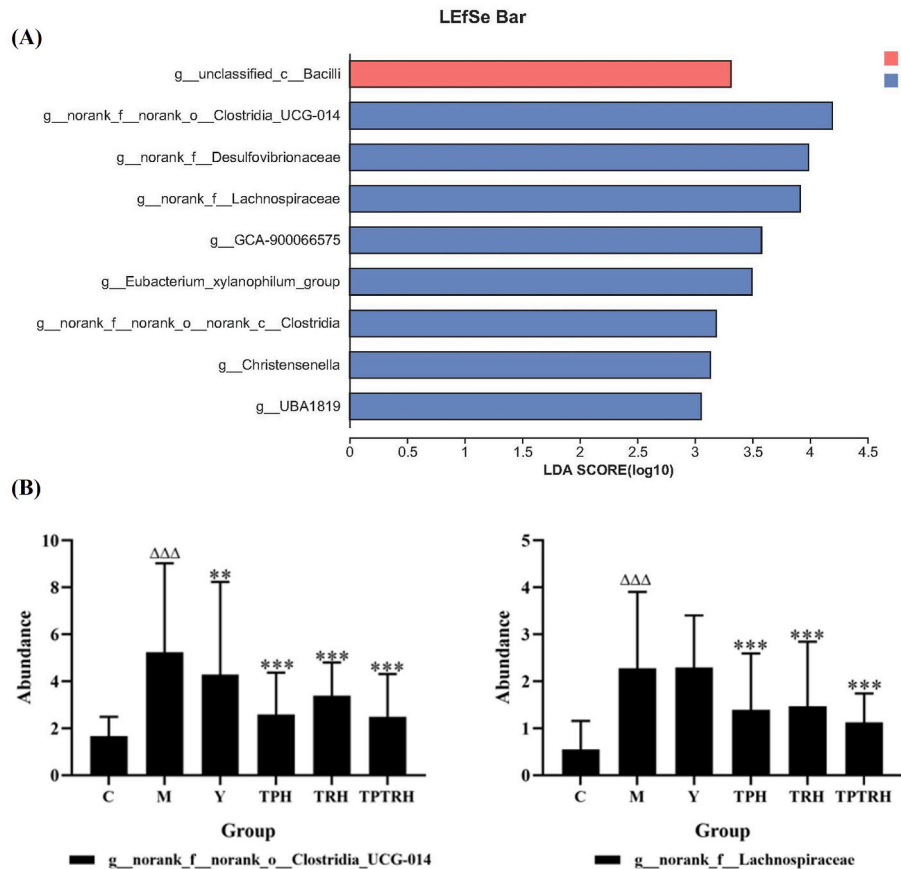


Fig. 4. Analysis of differential bacteria in cough mice (n = 6). C: the normal control group. M: the model control group. Y: the positive drug group. TPH: TP high dose group. TRH: TR high dose group. TPTRH: TP-TR high dose group. $\Delta\Delta\Delta P < 0.001$, vs. the normal control group. $**P < 0.01$, $***P < 0.001$, vs. the model control group.

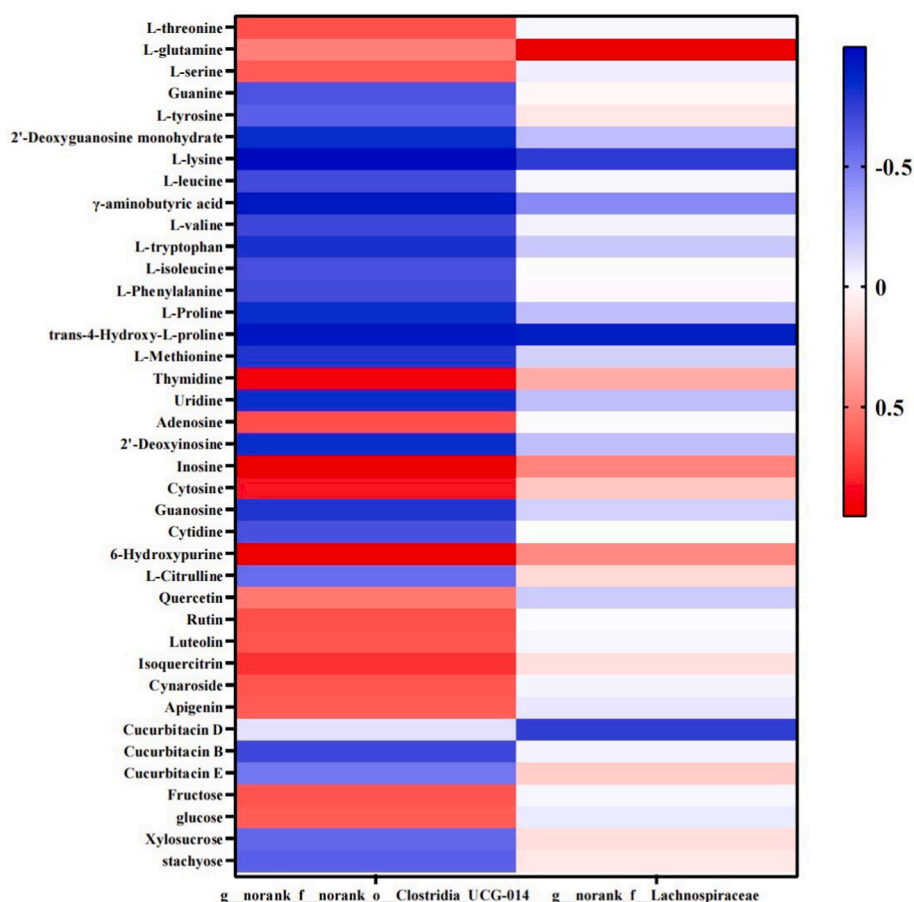


Fig. 5. Heat map of correlation coefficient between intestinal flora index and component measurement index of TP-TR used to intervene cough mice.

composition at the phylum level revealed that the predominant intestinal microbes in the mice belonged to the phyla *Firmicutes*, *Bacteroidetes*, *Actinobacteria*, and *Desulfobacteria*.

The alpha diversity of samples (Chao 1 index, Shannon index, Simpson index and ace index) indicated that the microflora had a high richness in samples. A high Shannon value and a low Simpson value can show higher species diversity. Compared with the control group, the Ace value, Chao value and Shannon value of the model group were found increased, while the Simpson value was decreased. The indexes in each administration group were returned to the level of the control group (supplementary materials Figs. S5–B).

The LEfSe analysis was used to identify the significant differences between the model group and the control group. Compared with the control group, the abundance of *Clostridia_UCG-014*, *desulfoaevibrionaceae*, *Lachnospiraceae*, *g_GCA-900066575*, *g_Eubacterium_xylanophilum_group*, *Clostridia* and *Christensenella* were significantly up-regulated and *bacilli* was significantly down-regulated in the model group (Fig. 4A). By COG and KEGG functions of mice intestinal flora prediction analysis, data showed that the disturbance of intestinal flora in cough mice may induce variation in some metabolic pathways, and the abundance of secondary metabolic functions, such as amino acid transport and metabolism, carbohydrate transport and metabolism, energy production and conversion, membrane transport, nucleotide metabolism decreased to the normal after administration of TP - TR (supplementary materials Fig. S6).

In addition, the abundance of *Clostridia_UCG-014* in the TP - TR group decreased from 5.24% (model group) to 2.48%, and *Lachnospiraceae* decreased from 2.28% (model group) to 1.12% (Fig. 4B, supplementary materials Fig. S7). Therefore, we correlated the abundance of these two strains with the content of TP-TR components. The results

showed that *Clostridia_UCG-014* was closely related to 2'-Deoxyguanosine monohydrate, L-lysine, L-leucine, γ-aminobutyric acid, L-valine, L-tryptophan, L-proline, trans-4-hydroxy-L-proline, L-methionine, uridine, 2'-deoxyinosine, guanosine, and cucurbitacin B ($|r| > 0.7$), while *Lachnospiraceae* was closely related to L-lysine ($r = -0.771$), trans-4-hydroxy-L-proline ($r = -0.917$) and cucurbitacin D ($r = -0.756$) (Fig. 5). This data suggested that the components may be the material basis of TP-TR to regulate intestinal flora in cough mice.

4. Discussion

The combination of *Trichosanthis Pericarpium* and *Trichosanthis Radix* offers a dual benefit in relieving cough symptoms. It not only possesses heat-clearing and phlegm-resolving properties but also promotes body fluid production and alleviates dryness. This combination has been observed to effectively alleviate cough symptoms. However, the specific proportions and material basis for the compatibility of these two herbs have not been clearly determined, although Mr. Shi Jinmo has remarked on the doubled cough-relieving effect achieved through their combination.

The effectiveness of TCM prescriptions heavily relies on their compatibility, and understanding the correlation between the material basis and biological effects serves as a significant point of entry in studying the mechanism of compatibility. In this study, a zebrafish inflammatory model combined with response surface analysis demonstrated that the 1:2 combination ratio of TP-TR had superior effects. Notably, during the co-decocting process of TP and TR, there were interactions among the chemical components. The TP-TR combination (1:2) exhibited enhanced dissolution of chemical components such as glutamine, tyrosine, citriline, thymidine, inosine, cytosine,

hypoxanthine, and isoquercetin. Previous studies have highlighted the crucial roles of nucleosides and amino acids in regulating inflammation, while glutamine supplementation has shown to reduce the serum levels of IL-1 β , TNF- α , and other inflammatory factors in patients with respiratory tract infections (Mohajeri et al., 2022). Inosine, as a source of physiological energy, has also been reported to have anti-inflammatory and immunological effects. It has been reported that inosine pretreatment can reduce serum levels of IL-6, IL-1 β and TNF- α , and effectively relieve LPs-induced inflammation (Guo et al., 2021). Quercetin, isoquercetin and other flavonoids can undergo extensive metabolism after entering human body due to their specific parent nucleus structure, which can closely interact with intestinal flora (Duda-Chodak et al., 2015). Moreover, Flavonoids, including quercetin, possess the ability to modulate the dominant bacteria in the intestine by reducing the ratio of Firmicutes to Bacteroides at the phylum level. Additionally, they can stimulate the production of short-chain fatty acids by the intestinal flora, thereby contributing to the maintenance of the body's immune system (Eteberria et al., 2015; Hidalgo et al., 2012). These findings may explain the rationality of TP-TR compatibility from the perspective of components function.

Cough is a natural defense mechanism that clears bronchial secretions and inhaled foreign particles (Morice and Shanks, 2017). The occurrence and development of cough are closely associated with the inflammatory response in the airways. Concentrated ammonia induced cough is a classic model for studying dry cough response, where the body produces a stress response to ammonia and induces inflammation (Mao et al., 2022; Zhang et al., 2018). Research indicates that during the inflammatory immune response, various sensitizing substances, including pro-inflammatory factors and chemokines such as IL-6, TNF- α , IL-1 β , CXCL1, CXCL5, and others, are released (O'Neill et al., 2013). The immune system of zebrafish is highly conserved, and tail cutting treatment can induce local damage to the tail of zebrafish, leading to immune response (Martin and Feng, 2009). Both as a stress response, here, we used the ammonia induced cough mice model to validate the optimal ratio of TP-TR screened from the zebrafish inflammation model, and the network pharmacology results. The results from the network pharmacology showed that IL-6, TNF, IL1 β , EGFR, AKT1 and other inflammatory-related factors played a crucial role in the treatment of cough with TP - TR (1:2), which was consistent with the results of the cough mice validation experiment.

Studies have shown that EGFR activation can increase TNF- α , IL-1 β expression that promote inflammatory damage, while EGFR inactivation is known to inhibit NF- κ B nuclear translocation, thereby significantly reducing the production of pro-inflammatory factors (Li et al., 2021). TLR4 upregulate the expression of various inflammatory factors by activating the NF- κ B pathway, thus exacerbating cough symptoms and leading to further deterioration of trachea and lung damage. STAT3 activation was also found to be a key signaling mediator of TLR4-induced glycolysis, macrophage metabolic reprogramming, and inflammation (Balic et al., 2020). In addition, cytokines such as IL-6, IL-1 β , and TNF- α also play an important role in the inflammatory response (Hirano, 2021) within a positive correlation between the activation of AKT1 and IL-6 expression and STAT3 phosphorylation (Malanga et al., 2015). In the cough model here, the expression of the above inflammatory factors was upregulated, and the mutual change trend of the factors was consistent with previous reports. The TP -TR herbal pair intervention had a callback effect on them, indicating that the compatibility reduced the inflammatory response caused by cough.

Gut microbiota is closely related to human health, and there is substantial evidence that gut microbial perturbations are associated with immunity and pulmonary airway inflammation (Osei Sekyere et al., 2020). Cough is a common symptom in respiratory diseases, prolonged cough can damage the lungs. In TCM, lung being considered to connected with large intestine. Here, Clostridia UCG-014, Lachnospiraceae, Ruminococcus and so on in the cough mice intestinal tract was found significantly higher compared to the normal mice. The TP-TR can

effectively reduce the abundance of Clostridia UCG-014 and Lachnospiraceae, suggesting that the two strains may be the target of herbal pair action. Few Clostridium genera, such as *Clostridium perfringens*, can cause inflammation, oxidative stress and cell damage leading to disease (Lee et al., 2020). Lachnospiraceae is also reported to increase inflammation in the body as a pro-inflammatory signal (Xie et al., 2016). Moreover, the correlation analysis revealed a strong association between the two strains and the presence of 2'-Deoxyguanosine monohydrate, L-lysine, L-leucine, γ -aminobutyric acid, L-valine, L-tryptophan, L-proline, trans-4-hydroxy-L-proline, L-methionine, uridine, 2'-deoxyinosine, guanosine, cucurbitacin B, and cucurbitacin D in TP-TR. Previous studies have reported that amino acids possess protective properties for intestinal epithelial cells, and they can effectively reduce intestinal inflammation and oxidative stress (He et al., 2018). The production of some pro-inflammatory factors, such as IL-8 and TNF- α , is also regulated by L-lysine (Zhang, Y. et al., 2019). Modern pharmacological studies have confirmed that cucurbitacin B has significant therapeutic effects on inflammatory diseases (Dai et al., 2023). Based on these findings, it can be hypothesized that the anti-inflammatory properties of TP-TR (1:2) is attributed to the presence of amino acids, nucleosides and cucurbitacins. It is likely that the regulation of intestinal flora contributes to the anti-inflammatory effects observed.

This study still has several limitations. Firstly, though we identified 2'-Deoxyguanosine monohydrate, L-lysine, L-leucine, γ -aminobutyric acid, L-valine, L-tryptophan, L-proline, trans-4-hydroxy-L-proline, L-methionine, uridine, 2'-deoxyinosine, guanosine, cucurbitacin B and cucurbitacin D as probable key chemicals in TP-TR, they may not be equivalent to this herbal pair. Usually, herbal pair compatibility plays a therapeutic role in the formulae, so whether the usage of reasonable combination doses of these compounds has similar effects remains to be confirmed. Secondly, cough is an important defense reflex of the respiratory tract, and it is also a common symptom of many respiratory disorders such as asthma, chronic bronchitis and COPD. In the network pharmacological analysis, several pathways such as HIF-1 signaling pathway, Toll-like receptor signaling pathway and so on were enriched. We only selected representative inflammatory factors for experimental verification, but not involved predicted target genes on other pathways. Furthermore, although we screened two strains of intestinal bacteria associated with cough, how TP-TR interacts with them still needs to be experimental confirmed.

5. Conclusions

To summarize, TP-TR in a ratio of 1:2 exhibited a superior anti-inflammatory effect. The compatibility of TP and TR facilitated the dissolution of components such as glutamine, inosine, and isoquercetin. EGFR, TLR4, AKT1, and STAT3 were confirmed as the regulatory targets for TP-TR in cough treatment. Furthermore, TP-TR demonstrated the ability to modulate the composition of the intestinal flora in cough-afflicted mice. The active ingredients responsible for its anti-inflammatory properties were identified as amino acids, nucleosides and cucurbitacins.

CRedit authorship contribution statement

Pei Liu: Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Xiao-ying Tan:** Validation, Formal analysis, Data curation, Writing – original draft. **Huang-qin Zhang:** Validation, Formal analysis, Data curation, Funding acquisition. **Ke-lei Su:** Investigation, Resources. **Er-xin Shang:** Methodology, Software, Validation. **Qing-ling Xiao:** Visualization, Supervision. **Sheng Guo:** Methodology, Investigation, Resources. **Jin-ao Duan:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2023.117096>.

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